Access control lists and group-based permissions may seem straightforward, but they become increasingly opaque in large, distributed organizations. In our multi-national retailer example, the U.S. support agent's access to German customer data might result from layered group memberships—perhaps the agent is in a general Support group that inherits permissions through nested regional or functional roles. These types of configurations are challenging to analyze or simulate. Answering basic questions like "Who has access to European customer data?" or "What would happen if we removed this group?" requires untangling group hierarchies or manually inspecting ACLs scattered across multiple systems. Consequently, policy drift is frequent, changes are risky, and even well-meaning updates can violate regulatory or business requirements without anyone noticing it.

1.2.8 Dynamic authorization addresses these challenges

The answer to these challenges is dynamic authorization using policies written in a machine-readable policy language. Policies can provide the scalability, flexibility, and other factors that modern applications and services need to meet the dynamic nature of today's digital landscape, ensuring secure and efficient access management across diverse environments.

1.3 Authorization matters

Most developers, product managers, technical managers, and business leaders are still largely unfamiliar with PBAC and its role in dynamic authorization, but this is rapidly changing. Several trends make dynamic authorization a more important topic than it has been.

1.3.1 Software as a Service

More and more software is being delivered as a service (so-called Software as a Service, or SaaS). This is especially true in certain categories like customer relationship management (CRM) and human resource management (HRM) where market penetration is estimated to be as high as 80%. Businesses, large and small, enjoy the ease of deployment, cost efficiency,